On the Theory of Modules

By *Ernst Steinitz* in Charlottenburg.

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The following investigations concern those modules whose introduction into number theory is due to *Dedekind*. In the eleventh supplement to *Dirichlet*'s lectures edited by him, the theory of modules is presented in its basic features. In addition, two further papers must be considered here: *Frobenius*, Theory of linear forms with integral coefficients; *Frobenius* and *Stickelberger*, On groups of interchangeable elements (Journal für die reine und angewandte Mathematik, vols. 86, 88). In the latter, the connection between modules and groups of interchangeable elements is made clear, and after this it is easy to transfer results obtained in one theory to the other. I shall therefore confine myself to the treatment of modules, although it would be natural to include group-theoretic questions.

I first state the most important relevant theorems of *Dedekind* and *Frobenius*, in the form and with the notation which I shall use throughout what follows.

§ 1. Rectangular systems; their division into classes.

1. Rectangular systems.

In what follows, capital Latin letters denote systems of rational integers arranged in rows and columns.[1] On such systems the operations addition, subtraction, and multiplication are to be applied in the usual way. Addition and subtraction are possible when the rectangular systems contain the same number of rows and the same number of columns; multiplication is possible when the number of columns of each factor is equal to the number of rows of the following factor, and the product then has as many rows as its first factor and as many columns as its last factor.

The number of columns of a rectangular system A is called its degree. If this degree is n and $0 < k \leq n$, then the greatest common divisor of the determinants of degree k in A is called "the kth determinant-divisor of A". It is also to be noted that the greatest common divisor of given numbers is taken positive and is to be put equal to 0 only when all the numbers are zero. If the number r of rows of A is $< n$, then nevertheless n determinant-divisors are to be counted, by stipulating that for $k > r$ the kth determinant-divisor is to be put equal to 0. The number of non-zero determinant-divisors of a rectangle is called its "rank".

If the system A is square, it is called a diagonal system when all elements outside the diagonal vanish. If $a_1, a_2, \ldots, a_n$ are the diagonal elements of such a system, it will also be denoted by $\mathrm{Dg}(a_1 \mid a_2 \mid \cdots \mid a_n)$. This diagonal system is called a "principal system" when none of the elements a is negative and each is a divisor of the following one. A square system A is called unimodular when its "norm" ($\mathrm{Nm}\, A$), that is, the absolute value of its determinant, is equal to 1. The letter E will always denote only unimodular systems; the symbol E_+ will denote those of determinant $+1$. If E^0 is the diagonal system of degree n all of whose diagonal elements are equal to 1, then for every square system A of degree n one has $AE^0 = A$, $E^0A = A$, and to every unimodular system E there belongs a second one E^{-1} for which

$$EE^{-1} = E^{-1}E = E^0$$

[1]The assumption that the elements are rational integers will be maintained throughout.

holds.

2. Division of square systems into classes.

We shall first restrict ourselves to considering square systems.

Definition 1. If A and B are square systems of the same degree n, and if A can be represented as a product in which B occurs as a factor, then we say: "The class of the system A ($\mathrm{Cl}\,A$) is divisible by the class of the system B."

If $\mathrm{Cl}\,A$ is divisible by $\mathrm{Cl}\,B$ and $\mathrm{Cl}\,B$ is divisible by $\mathrm{Cl}\,C$, then, as is immediately seen, $\mathrm{Cl}\,A$ is also divisible by $\mathrm{Cl}\,C$.

Definition 2. Two classes are called equal when each is divisible by the other.

The admissibility of these definitions is easily verified. The square systems are thereby divided into classes; each class is completely determined by a single system contained in it.

The problems now arise: given two square systems A and B, decide whether $\mathrm{Cl}\,A$ is divisible by $\mathrm{Cl}\,B$, or whether the systems A and B belong to the same class. These questions are decided by two theorems, probably first developed by *Frobenius*, which in what follows will be called Fundamental Theorems I and II.

Fundamental Theorem I. In order that two square systems A and B belong to the same class, it is necessary and sufficient that they possess the same determinant-divisors. If this requirement is fulfilled, A can always be represented in the form

$$A = EBE'$$

and, if in addition the determinants of A and B agree also in sign, also in the form

$$A = E_+BE'_+.$$

To this theorem there is joined a second one, which at the same time reveals an important property of the determinant-divisors.

Ia. Every class contains one and only one principal system.

Let $\mathrm{Dg}(e_1 \mid e_2 \mid \cdots \mid e_n)$ be a principal system, and let $d_1, d_2, \ldots, d_n$ be its determinant-divisors. Then $d_k = e_1e_2\cdots e_k$. Hence, if $d_1, d_2, \ldots, d_n$ are the determinant-divisors of any square system and the first r of them are non-zero, then

$$e_1 = d_1, \qquad e_2 = \frac{d_2}{d_1}, \quad \ldots, \quad e_r = \frac{d_r}{d_{r-1}}, \qquad e_{r+1} = 0, \ldots, e_n = 0$$

are the elements of the principal system of its class; hence, among the quotients

$$\frac{d_1}{1}, \quad \frac{d_2}{d_1}, \quad \ldots, \quad \frac{d_r}{d_{r-1}},$$

each is a divisor of the following one.

The diagonal elements of the principal system are called the "elementary divisors" or "invariants" of its class, or also of every system of this class. In what follows classes are denoted by capital Greek letters, or by giving their invariants; thus, if $a_1, a_2, \ldots, a_n$ are the invariants of $\mathrm{Cl}\,A = A$, we write

$$A = \mathrm{Cl}(a_1 \mid a_2 \mid \cdots \mid a_n).$$

All systems of a class A agree in degree, rank, norm, and determinant-divisors, and consequently one can speak of the degree, rank, and so on of the class A.

Fundamental Theorem II. The class A is divisible by the class B if and only if each invariant of A is divisible by the corresponding invariant of B.

To simplify expression, we now introduce a few elementary notations. The greatest common divisor of s numbers $a_1, a_2, \ldots, a_s$ will be denoted by

$$(a_1, a_2, \ldots, a_s),$$

and their least common multiple by

$$[a_1, a_2, \ldots, a_s].$$

The latter is to be put equal to 0 as soon as one of the numbers a vanishes; otherwise it is to be taken positive. Now let $A_1, A_2, \ldots, A_s$ be classes of degree n, let

$$A_i = \mathrm{Cl}(a_{i1} \mid a_{i2} \mid \cdots \mid a_{in}) \qquad (i = 1, \ldots, s),$$

and put

$$(a_{1k}, a_{2k}, \ldots, a_{sk}) = d_k, \qquad [a_{1k}, a_{2k}, \ldots, a_{sk}] = c_k \qquad (k = 1, \ldots, n).$$

Then $d_1, d_2, \ldots, d_n$ form the system of invariants of a certain class of degree n, and the same is true of the numbers $c_1, c_2, \ldots, c_n$. The first of these classes is denoted by

$$(A_1, A_2, \ldots, A_s),$$

the second by

$$[A_1, A_2, \ldots, A_s].$$

Thus $(A_1, A_2, \ldots, A_s)$ is the greatest common divisor, and $[A_1, A_2, \ldots, A_s]$ the least common multiple, of the classes $A_1, A_2, \ldots, A_s$. That is, the divisors common to all the classes $A_1, \ldots, A_s$ are exactly the divisors of $(A_1, \ldots, A_s)$, and the multiples common to all the classes $A_1, A_2, \ldots, A_s$ are exactly the multiples of $[A_1, \ldots, A_s]$. The class of degree n whose invariants are all equal to 1, and which is therefore a divisor of every class of degree n, will be denoted by $E^{(n)}$ (or simply by E when the degree is clear from the context). Two classes A, B are called relatively prime when $(A, B) = E$.

3. Classes of positive norm.

Let $A = \mathrm{Cl}(a_1 \mid a_2 \mid \cdots \mid a_n)$ be a class of positive norm, and let $\mathrm{Nm}\, A = a$ be the product of the relatively prime numbers a' and a''. Then A has only one divisor A' of norm a' and only one divisor A'' of norm a''. These classes

$$A' = \mathrm{Cl}((a_1, a') \mid (a_2, a') \mid \cdots \mid (a_n, a')),$$

$$A'' = \mathrm{Cl}((a_1, a'') \mid (a_2, a'') \mid \cdots \mid (a_n, a''))$$

are called "components" of A; A is "decomposable" into the components A' and A''. The characteristic relations for this decomposability are

$$(A', A'') = E, \qquad [A', A''] = A.$$

The decomposition of A can be continued until one reaches primary classes, i.e. classes whose norm is a power of a prime. In general, if m is a positive integer and p a prime, let m_p denote the highest power of p dividing m; correspondingly put

$$A_p = \mathrm{Cl}((a_1)_p \mid (a_2)_p \mid \cdots \mid (a_n)_p).$$

Then A_p is the primary component of A corresponding to the prime p. For primes p which do not divide $\operatorname{Nm} A$, one has $A_p = E$. Every class A of positive norm can be decomposed into primary components in one way only, and is the least common multiple of these components. If A is divisible by B, then for every prime p, A_p is divisible by B_p; and if for every prime p, A_p is divisible by B_p, then A is divisible by B. If $A_1, A_2, \ldots, A_s, \Delta, \Gamma$ are classes of positive norm, then the relations

$$(A_1, A_2, \ldots, A_s) = \Delta, \qquad [A_1, A_2, \ldots, A_s] = \Gamma$$

are respectively equivalent to the systems of relations

$$((A_1)_p, (A_2)_p, \ldots, (A_s)_p) = \Delta_p, \qquad [(A_1)_p, (A_2)_p, \ldots, (A_s)_p] = \Gamma_p.$$

4. Division of rectangular systems into classes.

In order now to include all rectangular systems among the classes, stipulate that every rectangular system A belongs to the class which possesses the same determinant-divisors as A. If the invariants of the class $\operatorname{Cl} A$ are also called the invariants of the system A, then the following theorem holds:

If A and B are rectangular systems, then in order that an equation of the form

$$A = P \cdot B \cdot Q$$

be possible, it is necessary and sufficient that $\operatorname{Rg} A \leq \operatorname{Rg} B$ and that each positive invariant of A be divisible by the corresponding invariant of B.

If A and B have the same degree, this condition amounts to $\operatorname{Cl} A$ being divisible by $\operatorname{Cl} B$. If the rectangles A and B have the same degree (n) and the same number of rows (r), and if $\operatorname{Cl} A$ is divisible by $\operatorname{Cl} B$, then in an equation of the form $A = P \cdot B \cdot Q$ the systems P and Q are square systems; and if $\operatorname{Cl} A = \operatorname{Cl} B$, it can always be arranged that P and Q are unimodular.

§ 2. Modules.

1. Modules.

According to *Dedekind*, a system of numbers is called a “module” if the difference of any two numbers of the system again belongs to the system. The module $\mathfrak{a}$ is said to be divisible by the module $\mathfrak{b}$ (a multiple of $\mathfrak{b}$, and $\mathfrak{b}$ a divisor of $\mathfrak{a}$) when every number of $\mathfrak{a}$ is also contained in $\mathfrak{b}$. If the numbers

$$\alpha_1, \alpha_2, \ldots, \alpha_r$$

belong to the module $\mathfrak{a}$, then the same is true of every number of the form

$$\alpha = c_1\alpha_1 + c_2\alpha_2 + \cdots + c_r\alpha_r, \tag{1}$$

where the coefficients c are rational integers. The module $\mathfrak{a}$ is called finite when one can take from it a finite number of numbers $\alpha_1, \alpha_2, \ldots, \alpha_r$ such that every number of $\mathfrak{a}$ can be represented in the form (1). The numbers $\alpha_1, \alpha_2, \ldots, \alpha_r$ are then said to form a basis of $\mathfrak{a}$, and this will be indicated by writing

$$\mathfrak{a} = \operatorname{Md}(\alpha_1, \alpha_2, \ldots, \alpha_r).$$

The basis is called reduced if the expression (1) can vanish only by all coefficients vanishing. Then every number α of $\mathfrak{a}$ can be brought into the form (1) in only one way; every basis of $\mathfrak{a}$ consists of at least r numbers; and the module $\mathfrak{a}$ is called r-component. Every multiple of an r-component module is at most r-component.

Suppose one has an n-component module

$$\mathfrak{e} = \mathrm{Md}(\eta_1, \eta_2, \ldots, \eta_n).$$

If one considers only numbers of this module and performs no operations on them other than addition and subtraction, the basis elements $\eta_1, \eta_2, \ldots, \eta_n$ may be treated as indeterminate quantities. This amounts to considering, in place of the numbers contained in $\mathfrak{e}$

$$\eta = c_1\eta_1 + \cdots + c_n\eta_n,$$

the systems of the n coefficients

$$c_1, c_2, \ldots, c_n$$

by which the numbers η are assembled from the numbers $\eta_1, \eta_2, \ldots, \eta_n$. Such a system of coefficients will always be thought of as arranged in a horizontal row. Doing this, while otherwise retaining the notation used so far, we shall understand by a module a totality of systems or "rows" σ which reproduce themselves under subtraction.

Now let

$$\mathfrak{a} = \mathrm{Md}(\alpha_1, \alpha_2, \ldots, \alpha_r)$$

be any module. If one forms the rectangle A whose rows are $\alpha_1, \alpha_2, \ldots, \alpha_r$, then the module $\mathfrak{a}$ is completely determined by it. We call the rectangle A a basis of the module $\mathfrak{a}$ and write

$$\mathfrak{a} = \mathrm{Md}\, A.$$

The necessary and sufficient condition that the module $\mathfrak{b} = \mathrm{Md}\, B$ be divisible by the module $\mathfrak{a} = \mathrm{Md}\, A$ is evidently the existence of an equation of the form

$$B = P \cdot A.$$

Taking account of the definitions and theorems given in § 1, it follows that:

If the rectangles $A_1, A_2, \ldots$ are bases of one and the same module $\mathfrak{a}$, then they all belong to the same class. It is convenient to call this class the class of the module $\mathfrak{a}$, so that

$$\mathrm{Cl}\, A_1 = \mathrm{Cl}\, A_2 = \cdots = \mathrm{Cl}\, \mathfrak{a}.$$

By transferring to the modules belonging to a class the concepts previously introduced for the class itself, it is then clear what is to be understood by the degree, rank, norm of a module $\mathfrak{a}$ ($\mathrm{Gd}\,\mathfrak{a}, \mathrm{Rg}\,\mathfrak{a}, \mathrm{Nm}\,\mathfrak{a}$), and by its determinant-divisors and invariants. One sees that the rank of a module is the same thing as what was earlier called its number of components, so that a module of rank r is r-component. Divisibility can of course occur only between modules of the same degree.

I now state several important theorems from the theory of modules, which will be used frequently later.

I. If $\mathrm{Rg}\,\mathfrak{a} = r$, the module $\mathfrak{a}$ possesses a basis of k systems if and only if $k \geq r$. Since $\mathrm{Gd}\,\mathfrak{a} \geq \mathrm{Rg}\,\mathfrak{a}$, every module of degree n also possesses a basis of n systems, a square basis.

II. Every module $\mathfrak{a}$ possesses a square basis of the form AE, where A is the principal system of $\mathrm{Cl}\,\mathfrak{a}$. The class $E^{(n)}$ contains only one module; this module is a divisor of every module of degree n, and will be denoted by $\mathfrak{e}^{(n)}$ (or $\mathfrak{e}$). Every unimodular system of degree n is a basis of $\mathfrak{e}^{(n)}$. Theorem II can therefore also be formulated as follows:

IIa. If $\mathrm{Cl}\,\mathfrak{a} = A = \mathrm{Cl}(a_1 \mid a_2 \mid \cdots \mid a_n)$, one can give n rows $\alpha_1, \alpha_2, \ldots, \alpha_n$ such that

$$\mathrm{Md}(\alpha_1, \alpha_2, \ldots, \alpha_n) = \mathfrak{e}, \qquad \mathrm{Md}(a_1\alpha_1, a_2\alpha_2, \ldots, a_n\alpha_n) = \mathfrak{a}.$$

If $\mathrm{Rg}\,\mathfrak{a} = r$, then also $\mathfrak{a} = \mathrm{Md}(a_1\alpha_1, \ldots, a_r\alpha_r)$.

III. If the module $\mathfrak{a}$ is a multiple of $\mathfrak{b}$, then $\mathrm{Cl}\,\mathfrak{a}$ is a multiple of $\mathrm{Cl}\,\mathfrak{b}$; and if the class A is a multiple of B, then every module of A has at least one divisor within B, and every module of B has at least one multiple within A.

IV. If $\mathfrak{a}$ is a multiple of $\mathfrak{b}$ and $\mathrm{Cl}\,\mathfrak{a}$ is a divisor of $\mathrm{Cl}\,\mathfrak{b}$, then $\mathfrak{a} = \mathfrak{b}$.

Let $\mathfrak{a}_1 = \mathrm{Md}\,A_1$, $\mathfrak{a}_2 = \mathrm{Md}\,A_2$, $\ldots$, $\mathfrak{a}_s = \mathrm{Md}\,A_s$ be modules of the same degree n. They possess a greatest common divisor, that is, there exists a definite module $\mathfrak{b}$ with the property that every divisor common to the modules $\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s$ is a divisor of $\mathfrak{b}$, and conversely. The rectangle whose rows are all the rows of the rectangles $A_1, A_2, \ldots, A_s$ is a basis of this module. The modules $\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s$ also possess a least common multiple, that is, there exists a definite module $\mathfrak{c}$ with the property that every common multiple of the modules $\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s$ is a multiple of $\mathfrak{c}$, and conversely. Simple considerations show how to find a basis of this module $\mathfrak{c}$ by a finite number of operations, even in the cases in which some of the modules $\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s$ have vanishing norm. We denote by

$$(\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s), \qquad [\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s][2]$$

the greatest common divisor and the least common multiple of the modules $\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s$, respectively.

The fact that the module $\mathfrak{a}$ is a multiple of the module $\mathfrak{b}$ may be expressed by either of the equations

$$(\mathfrak{a}, \mathfrak{b}) = \mathfrak{b}, \qquad [\mathfrak{a}, \mathfrak{b}] = \mathfrak{a},$$

but also, since $(\mathfrak{a}, \mathfrak{b})$ is in any case a divisor of $\mathfrak{b}$, by Theorem IV through the equation

$$\mathrm{Cl}(\mathfrak{a}, \mathfrak{b}) = \mathrm{Cl}\,\mathfrak{b}.$$

This gives a simple means of recognizing whether, for two modules $\mathfrak{a}, \mathfrak{b}$ each given by a basis, the first has the second as divisor. For then one also has a basis of $(\mathfrak{a}, \mathfrak{b})$, so one can immediately determine the class of the module $(\mathfrak{a}, \mathfrak{b})$ and has only to see whether it is identical with $\mathrm{Cl}\,\mathfrak{b}$.

V. Between the norms of two modules $\mathfrak{a}, \mathfrak{b}$, their greatest common divisor, and their least common multiple, there is the relation

$$\mathrm{Nm}\,\mathfrak{a}\ \mathrm{Nm}\,\mathfrak{b} = \mathrm{Nm}(\mathfrak{a}, \mathfrak{b})\ \mathrm{Nm}[\mathfrak{a}, \mathfrak{b}].$$

If A is the principal system of the class A, the module $\mathrm{Md}\,A$ is called the "principal module" of the class A. In order that the class A be divisible by B, it is necessary and sufficient that the principal module of A be divisible by the principal module of B. The greatest common divisor (respectively, the least common multiple) of the principal modules of the classes $A_1, A_2, \ldots, A_s$ is the principal module of the class $(A_1, A_2, \ldots, A_s)$ (respectively, $[A_1, A_2, \ldots, A_s]$). Two modules $\mathfrak{a}, \mathfrak{b}$ are called relatively prime when $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{e}$.

2. Unimodular transformations.

Let A and B be square systems of degree n, let E be a unimodular system of the same degree, and suppose there is an equation

$$A = CB. \tag{2}$$

Then it follows that

$$AE = CBE. \tag{3}$$

Conversely, (2) also follows from (3). This result can be stated as follows:

[2]In Dedekind's notation, $\mathfrak{a}_1 + \mathfrak{a}_2 + \cdots + \mathfrak{a}_s$ and $\mathfrak{a}_1 - \mathfrak{a}_2 - \cdots - \mathfrak{a}_s$, respectively.

If $\operatorname{Md} A$ is divisible by $\operatorname{Md} B$, then $\operatorname{Md} AE$ is divisible by $\operatorname{Md} BE$, and conversely. Of the relations

$$\operatorname{Md} A = \operatorname{Md} B, \qquad \operatorname{Md} AE = \operatorname{Md} BE, \tag{4}$$

each is therefore a consequence of the other. Thus, for every square basis A of a definite module $\mathfrak{a}$, AE represents the basis of a definite second module $\mathfrak{a}'$. We therefore say:

"The module $\mathfrak{a}$ is carried by the unimodular transformation $[E]$ into the module $\mathfrak{a}'$"

and write

$$\mathfrak{a}' = \mathfrak{a}E.$$

The inverse transformation $[E^{-1}]$ carries $\mathfrak{a}'$ back into $\mathfrak{a}$.

From the equivalence of equations (2) and (3) it follows further that:

VI. If the module $\mathfrak{a}$ is divisible by the module $\mathfrak{b}$, then $\mathfrak{a}E$ is divisible by $\mathfrak{b}E$, and conversely.

A simple consideration then gives:

VII. The greatest common divisor (respectively, the least common multiple) of the modules $\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s$ is carried by the unimodular transformation $[E]$ into the greatest common divisor (respectively, the least common multiple) of the transformed modules; in other words, the identities

$$(\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s)E = (\mathfrak{a}_1 E, \mathfrak{a}_2 E, \ldots, \mathfrak{a}_s E),$$

$$[\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s]E = [\mathfrak{a}_1 E, \mathfrak{a}_2 E, \ldots, \mathfrak{a}_s E]$$

hold.

If A is a square basis of the module $\mathfrak{a}$ and $\mathfrak{a}E = \mathfrak{a}'$, then $\mathfrak{a}' = \operatorname{Md} AE$ and $\operatorname{Cl} \mathfrak{a}' = \operatorname{Cl} AE = \operatorname{Cl} A = \operatorname{Cl} \mathfrak{a}$. Conversely, if $\mathfrak{a}' = \operatorname{Md} A'$ is a module of the class $\operatorname{Cl} \mathfrak{a}$ and A' is square, then by Fundamental Theorem I one may write $A' = E_1 A E_2$, and it follows that

$$\mathfrak{a}' = \operatorname{Md} E_1 A E_2 = \operatorname{Md} A E_2 = \mathfrak{a} E_2.$$

Thus:

VIII. A module $\mathfrak{a}$ can be carried by unimodular transformations into every module of its class, but into no further module.

This is the basis for the fact that all essential properties of a module are determined by its class.

One of the most important problems to be solved will consist in determining the number of modules which belong to the same class A, given by its system of invariants. We denote this number, which is a "function of A", by

$$\psi(A).$$

It is easy to see that for classes whose norm vanishes, $\psi(A) = \infty$, except in the case in which all invariants of A vanish, where evidently one must put $\psi(A) = 1$. We shall therefore be able to restrict ourselves to determining the function ψ for classes of positive norm.

3. Multiplication of modules by a number.

Let $\alpha_1, \alpha_2, \ldots, \alpha_r$ be rows of n integers each, and let m be any integer. Then the module $\operatorname{Md}(m\alpha_1, m\alpha_2, \ldots, m\alpha_r)$ consists of all rows which arise from the rows of the module $\operatorname{Md}(\alpha_1, \alpha_2, \ldots, \alpha_r)$ by multiplication by m. We denote it by "$m \cdot \mathfrak{a}$" and suppose $m > 0$. If $\mathfrak{a}$ is divisible by $\mathfrak{b}$, then $m\mathfrak{a}$ is divisible by $m\mathfrak{b}$, and conversely. If the module $\mathfrak{a}$ has invariants $a_1, a_2, \ldots, a_n$, then $m\mathfrak{a}$ has invariants $ma_1, ma_2, \ldots, ma_n$; conversely, if a module $\mathfrak{c}$ has invariants $ma_1, ma_2, \ldots, ma_n$, one may write $\mathfrak{c} = m\mathfrak{a}'$, and then $a_1, a_2, \ldots, a_n$ are the invariants of $\mathfrak{a}'$.

Hence, if $A = \mathrm{Cl}(a_1 \mid a_2 \mid \cdots \mid a_n)$ is any class and if $\mathrm{Cl}(ma_1 \mid ma_2 \mid \cdots \mid ma_n) = mA$ is put, the classes A and mA contain equally many modules, i.e.

$$\text{IX.} \qquad \psi(mA) = \psi(A).$$

The class E contains only the module $\mathfrak{e}$, and hence mE contains only the single module $m\mathfrak{e}$, so that for every unimodular system E one has $m\mathfrak{e}E = m\mathfrak{e}$.

X. In order that the module $\mathfrak{a}$ be a divisor or a multiple of $m\mathfrak{e}$, it is necessary (Theorem III) and sufficient that $\mathrm{Cl}\,\mathfrak{a} = A$ be a divisor or multiple of mE.

Indeed, if the stated condition is fulfilled, the principal module $\mathfrak{a}'$ of A is a divisor or multiple of $m\mathfrak{e}$. By Theorem VIII one can then write $\mathfrak{a} = \mathfrak{a}'E$, whence it follows that $\mathfrak{a}$ is a divisor or multiple of $m\mathfrak{e}E = m\mathfrak{e}$. In entirely similar fashion one proves:

XI. If A is any class of the same degree as E, and $\mathfrak{a}$ is any module of A, then

$$\mathrm{Cl}(\mathfrak{a}, m\mathfrak{e}) = (A, mE), \qquad \mathrm{Cl}[\mathfrak{a}, m\mathfrak{e}] = [A, mE].$$

4. Modules of positive norm.

XII. Let $\mathfrak{a}$ be a module of the class A, let $\mathrm{Nm}\,\mathfrak{a} = \mathrm{Nm}\,A = a > 0$, let a be the product of the relatively prime numbers a' and a'', and let A' and A'' be the components of A whose norms are a' and a'', respectively. Then $\mathfrak{a}$ has one and only one divisor $\mathfrak{a}'$ in the class A', and likewise one and only one divisor $\mathfrak{a}''$ in the class A'', and

$$\mathfrak{a} = [\mathfrak{a}', \mathfrak{a}''].$$

XIII. If the classes of the modules $\mathfrak{a}'$ and $\mathfrak{a}''$ are relatively prime, then

$$\mathrm{Cl}[\mathfrak{a}', \mathfrak{a}''] = [\mathrm{Cl}\,\mathfrak{a}', \mathrm{Cl}\,\mathfrak{a}''].$$

Proofs. That, under the hypotheses of XII, $\mathfrak{a}$ has a divisor $\mathfrak{a}'$ in the class A' is already asserted by Theorem III. We now show that $\mathfrak{a}$ has no divisor of norm a' other than $\mathfrak{a}'$. Suppose there were a second such module $\mathfrak{a}'_1$. Then $\mathfrak{a}'$ and $\mathfrak{a}'_1$ would be divisors of $\mathfrak{a}$ and, by Theorem X, of $a'\mathfrak{e}$; hence $[\mathfrak{a}', \mathfrak{a}'_1]$ would be a divisor of $(\mathfrak{a}, a'\mathfrak{e})$, $\mathrm{Cl}[\mathfrak{a}', \mathfrak{a}'_1]$ would be a divisor of $\mathrm{Cl}(\mathfrak{a}, a'\mathfrak{e}) = (A, a'E) = A' = \mathrm{Cl}\,\mathfrak{a}'$, and Theorem IV would imply $[\mathfrak{a}', \mathfrak{a}'_1] = \mathfrak{a}'$, i.e. $\mathfrak{a}'_1$ is a divisor of $\mathfrak{a}'$. Since, however, the norms of $\mathfrak{a}'$ and $\mathfrak{a}'_1$ are positive and equal, it follows that $\mathfrak{a}'_1 = \mathfrak{a}'$. Thus $\mathfrak{a}$ has only one divisor $\mathfrak{a}'$ in A', and likewise only one divisor $\mathfrak{a}''$ in A''.

From the fact that the classes $A' = \mathrm{Cl}\,\mathfrak{a}'$ and $A'' = \mathrm{Cl}\,\mathfrak{a}''$ are relatively prime, it already follows that $[\mathfrak{a}', \mathfrak{a}'']$ belongs to the class $A = [A', A'']$. Namely, put $\mathrm{Cl}[\mathfrak{a}', \mathfrak{a}''] = A_1$. Then $\mathrm{Nm}\,A_1 = \mathrm{Nm}[\mathfrak{a}', \mathfrak{a}'']$ is divisible by $[a', a''] = a$, while on the other hand, by Theorem V,

$$\mathrm{Nm}\,A_1 = \mathrm{Nm}[\mathfrak{a}', \mathfrak{a}''] = \frac{a'a''}{\mathrm{Nm}(\mathfrak{a}', \mathfrak{a}'')} = \frac{a}{\mathrm{Nm}(\mathfrak{a}', \mathfrak{a}'')}$$

is a divisor of a, so $\mathrm{Nm}\,A_1 = a$. The modules $\mathfrak{a}'$ and $\mathfrak{a}''$, which by the preceding are uniquely determined by the conditions that they have norms a' and a'' and are divisors of $[\mathfrak{a}', \mathfrak{a}'']$, must therefore belong to the components A'_1, A''_1 of A_1 satisfying $\mathrm{Nm}\,A'_1 = a'$, $\mathrm{Nm}\,A''_1 = a''$; and these can be identical with A', A'' only if also $A_1 = A$. Finally, since $\mathfrak{a}$ is divisible by $\mathfrak{a}'$ and $\mathfrak{a}''$, and hence by $[\mathfrak{a}', \mathfrak{a}'']$, and since $\mathrm{Cl}\,\mathfrak{a} = \mathrm{Cl}[\mathfrak{a}', \mathfrak{a}'']$, it follows that

$$\mathfrak{a} = [\mathfrak{a}', \mathfrak{a}''].$$

This proves Theorems XII and XIII.

XIV. Let the module $\mathfrak{a}$, of norm $a(>0)$, be divisible by the module $\mathfrak{b}$, of norm b. Let a be the product of the relatively prime numbers a' and a'', and let $\mathfrak{a}', \mathfrak{a}''$ be the divisors of $\mathfrak{a}$ which have norms a' and a''. Then the modules $\mathfrak{b}' = (\mathfrak{a}', \mathfrak{b})$ and $\mathfrak{b}'' = (\mathfrak{a}'', \mathfrak{b})$ have norms $b' = (a', b)$ and $b'' = (a'', b)$, and their least common multiple is $\mathfrak{b}$.

Proof.

$$\operatorname{Nm}[\mathfrak{a}', \mathfrak{b}] = \frac{a'b}{\operatorname{Nm}(\mathfrak{a}', \mathfrak{b})}$$

is a divisor of $a'b$ and also of $\operatorname{Nm}\mathfrak{a} = a = a'a''$, hence also of $a'(a'', b) = a'b''$. Therefore

$$\operatorname{Nm}\mathfrak{b}' = \operatorname{Nm}(\mathfrak{a}', \mathfrak{b}) = \frac{a'b}{\operatorname{Nm}[\mathfrak{a}', \mathfrak{b}]}$$

is divisible by $a'b/(a'b'') = b/b'' = b'$. Conversely, $\operatorname{Nm}\mathfrak{b}'$ is evidently a divisor of $(a', b) = b'$. Thus $\operatorname{Nm}\mathfrak{b}' = b'$, and similarly $\operatorname{Nm}\mathfrak{b}'' = b''$. Since b' and b'' are relatively prime, it follows from XII and XIII that $\operatorname{Nm}[\mathfrak{b}', \mathfrak{b}''] = [b', b''] = \operatorname{Nm}\mathfrak{b}$, and since $\mathfrak{b}$ is divisible by $[\mathfrak{b}', \mathfrak{b}'']$, it must now be that $[\mathfrak{b}', \mathfrak{b}''] = \mathfrak{b}$.

If $\mathfrak{a}'$ is a divisor of the module $\mathfrak{a}$ and $\operatorname{Cl}\mathfrak{a}' = A'$ is a component of $\operatorname{Cl}\mathfrak{a} = A$, we call $\mathfrak{a}'$ a "component of $\mathfrak{a}$". From XII and XIII it follows that each component of A corresponds to a definite component of $\mathfrak{a}$, and each decomposition of A into two components A', A'' corresponds to a decomposition of $\mathfrak{a}$ into two components $\mathfrak{a}'$ and $\mathfrak{a}''$ such that $[\mathfrak{a}', \mathfrak{a}''] = \mathfrak{a}$. If $\mathfrak{a}'$ is allowed to run through every module of A' and $\mathfrak{a}''$ through every module of A'', then $[\mathfrak{a}', \mathfrak{a}'']$ represents every module of A exactly once; hence the relation

$$\psi(A) = \psi(A') \cdot \psi(A'') \tag{5}$$

holds. The decomposition of the module $\mathfrak{a}$ can be continued until one reaches its primary components, whose norms are powers of primes. We denote by $\mathfrak{a}_p$ the component of $\mathfrak{a}$ belonging to the class A_p. From XIV one now obtains:

XV. If $\mathfrak{a}$ and $\mathfrak{b}$ are modules of positive norm, then for the divisibility of $\mathfrak{a}$ by $\mathfrak{b}$ it is necessary and sufficient that for every prime p, $\mathfrak{b}_p$ be a divisor of $\mathfrak{a}_p$.

It follows further that:

XVI. If $\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s$ are modules of positive norm, then the relations

$$(\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s) = \mathfrak{b}, \qquad [\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s] = \mathfrak{c}$$

are respectively equivalent to the systems of relations

$$((\mathfrak{a}_1)_p, (\mathfrak{a}_2)_p, \ldots, (\mathfrak{a}_s)_p) = \mathfrak{b}_p, \qquad [(\mathfrak{a}_1)_p, (\mathfrak{a}_2)_p, \ldots, (\mathfrak{a}_s)_p] = \mathfrak{c}_p.$$

Finally, from (5) one also obtains the formula

$$\text{XVII.} \qquad \psi(A) = \prod_p \psi(A_p),$$

where the product ranges over all primes, or only over those which divide $\operatorname{Nm} A$, since for the others $A_p = E$ and $\psi(A_p) = 1$.

§ 3. Congruences; the functions φ_r and φ.

Two rows σ, σ' each of n elements are called congruent modulo the module $\mathfrak{a}$ of degree n,

$$\sigma \equiv \sigma' \pmod{\mathfrak{a}},$$

when $\sigma - \sigma'$ belongs to the module $\mathfrak{a}$. If R, R' are rectangles of degree n with r rows each, then

$$R \equiv R' \pmod{\mathfrak{a}}$$

means that each row of R is congruent to the corresponding row of R'. Each row of n elements represents a definite residue of the module $\mathfrak{a}$, in the sense that congruent rows are regarded as representatives of the same residue. Likewise, a rectangle of degree n with r rows represents a definite r-rowed residue of $\mathfrak{a}$. From the congruence

$$R \equiv R' \pmod{\mathfrak{a}}$$

it follows that

$$RE \equiv R'E \pmod{\mathfrak{a}E},$$

where E denotes any unimodular system. Thus the unimodular transformation $[E]$, which carries $\mathfrak{a}$ into $\mathfrak{a}E$, at the same time carries every residue of $\mathfrak{a}$ into a definite residue of $\mathfrak{a}E$. Since the transformation is uniquely reversible, it establishes a one-to-one correspondence between the residues of $\mathfrak{a}$ and those of $\mathfrak{a}E$. From this and from § 2, VIII it follows that for all modules of a class, the number of residues of a specified row-number is the same. Consideration of the principal module of a class A shows that, if $\operatorname{Nm} A \neq 0$, the number of one-row residues of a module of A is equal to $\operatorname{Nm} A$, and consequently the number of r-rowed residues is equal to $(\operatorname{Nm} A)^r$. If, on the other hand, $\operatorname{Nm} A = 0$, then the number of residues is evidently infinite.

If to a basis of the module $\mathfrak{a}$ one adjoins the rows of a rectangle R, one obtains a basis of the module

$$(\mathfrak{a}, \operatorname{Md} R),$$

which, for brevity, we shall denote simply by

$$(\mathfrak{a}, R).$$

The module $(\mathfrak{a}, R)$ is a divisor of $\mathfrak{a}$. If $R \equiv R' \pmod{\mathfrak{a}}$, then

$$(\mathfrak{a}, R') = (\mathfrak{a}, R).$$

Now suppose that $\operatorname{Nm} \mathfrak{a} > 0$, and that

$$\mathfrak{a} = [\mathfrak{a}', \mathfrak{a}'']$$

is a decomposition of $\mathfrak{a}$ into components, so that $(\mathfrak{a}', \mathfrak{a}'') = \mathfrak{e}$. To every residue R of $\mathfrak{a}$ there then corresponds a definite residue R' of $\mathfrak{a}'$ and a residue R'' of $\mathfrak{a}''$, satisfying the congruences

$$R \equiv R' \pmod{\mathfrak{a}'}, \qquad R \equiv R'' \pmod{\mathfrak{a}''}. \tag{1}$$

Conversely, it is easily seen (for example, by comparing the numbers of residues of the modules $\mathfrak{a}, \mathfrak{a}', \mathfrak{a}''$) that to every combination of a residue of $\mathfrak{a}'$ and a residue of $\mathfrak{a}''$ with the same number of rows there belongs a residue R of $\mathfrak{a}$ determined by the congruences (1). We now prove:

I. If

$$\mathfrak{a} = [\mathfrak{a}', \mathfrak{a}'']$$

is a component decomposition of $\mathfrak{a}$, and if the residues R, R', R'' of $\mathfrak{a}, \mathfrak{a}', \mathfrak{a}''$ satisfy the congruences (1), then between the modules

$$\mathfrak{b} = (\mathfrak{a}, R), \qquad \mathfrak{b}' = (\mathfrak{a}', R'), \qquad \mathfrak{b}'' = (\mathfrak{a}'', R'')$$

there is the equality

$$\mathfrak{b} = [\mathfrak{b}', \mathfrak{b}''].$$

Proof. One has

$$\mathfrak{b}' = (\mathfrak{a}', R') = (\mathfrak{a}', R) = ((\mathfrak{a}, \mathfrak{a}'), R) = (\mathfrak{a}', (\mathfrak{a}, R)) = (\mathfrak{a}', \mathfrak{b}),$$

and similarly $\mathfrak{b}'' = (\mathfrak{a}'', \mathfrak{b})$. Hence by § 2, XIV, $[\mathfrak{b}', \mathfrak{b}''] = \mathfrak{b}$.

Let r be a positive integer, and let the number of r-rowed residues of $\mathfrak{a}$ for which

$$(\mathfrak{a}, R) = \mathfrak{e}$$

be denoted by

$$\varphi_r(\mathfrak{a}).$$

Since for every residue R and every unimodular system E the relation

$$(\mathfrak{a}E, RE) = (\mathfrak{a}, R)E$$

holds, the function φ_r has the same value for all modules of the same class. Thus $\varphi_r(\mathfrak{a})$ is a function only of the class of $\mathfrak{a}$; if $\mathrm{Cl}\,\mathfrak{a} = A$, we put

$$\varphi_r(\mathfrak{a}) = \varphi_r(A).$$

We now turn to the determination of the function $\varphi_r(A)$, excluding the case $\mathrm{Nm}\,A = 0$.

Again write

$$\mathfrak{a} = [\mathfrak{a}', \mathfrak{a}'']$$

as a component decomposition of $\mathfrak{a}$. Let R run through every r-rowed residue of $\mathfrak{a}$, and let R', R'' run through every combination of an r-rowed residue of $\mathfrak{a}'$ and one of $\mathfrak{a}''$, subject to the condition that R, R', R'' are connected by congruences (1). Then I shows that the equation

$$(\mathfrak{a}, R) = \mathfrak{e} \tag{2}$$

is equivalent to the system of equations

$$(\mathfrak{a}', R') = \mathfrak{e}, \qquad (\mathfrak{a}'', R'') = \mathfrak{e}. \tag{3}$$

Since (2) is satisfied by $\varphi_r(\mathfrak{a})$ residues R, and (3) by $\varphi_r(\mathfrak{a}') \cdot \varphi_r(\mathfrak{a}'')$ combinations R', R'', it follows that

$$\varphi_r(\mathfrak{a}) = \varphi_r(\mathfrak{a}') \cdot \varphi_r(\mathfrak{a}''),$$

or, putting $\mathrm{Cl}\,\mathfrak{a}' = A'$ and $\mathrm{Cl}\,\mathfrak{a}'' = A''$,

$$\varphi_r(A) = \varphi_r(A') \cdot \varphi_r(A'').$$

Continuing the decomposition of A into primary components gives

$$\text{II.} \qquad \varphi_r(A) = \prod_p \varphi_r(A_p).$$

Thus it remains only to determine the function φ_r for primary classes.

Let A be the principal system of the primary class $A = A_p$, so $\mathrm{Md}\,A = \mathfrak{a}$ is its principal module. Suppose that among the invariants of A only the last q are divisible by p, while the first $n - q$ are equal to 1. In order that, for an r-rowed rectangle R, one have $(\mathfrak{a}, R) = \mathfrak{e}$, the determinants of degree n in the rectangle formed from the rows of A and R must have 1 as greatest common divisor. Evidently this is possible only when $r \geq q$. Thus for $r < q$ one has $\varphi_r(A) = 0$. Now suppose $r \geq q$. In order that $(\mathfrak{a}, R) = \mathfrak{e}$, it is necessary and sufficient that the determinants of degree q which can be taken from the last q columns of R should not all be divisible by p. The module $(\mathfrak{a}, p\mathfrak{e})$ is the principal module of the class (A, pE), whose first $n - q$

invariants are equal to 1 and whose last q invariants are equal to p. If the rectangle R satisfies $(\mathfrak{a}, R) = \mathfrak{e}$, then also $(\mathfrak{a}, p\mathfrak{e}, R) = \mathfrak{e}$; conversely, from $(\mathfrak{a}, p\mathfrak{e}, R) = \mathfrak{e}$ it follows that $(\mathfrak{a}, R) = \mathfrak{e}$. Moreover, under the representatives of a definite r-rowed residue of $(\mathfrak{a}, p\mathfrak{e})$, there are exactly

$$\frac{\mathrm{Nm}\,\mathfrak{a}}{p^q}$$

residues incongruent modulo $\mathfrak{a}$. Hence one obtains the relation

$$\varphi_r(\mathfrak{a}) = \left(\frac{\mathrm{Nm}\,\mathfrak{a}}{p^q}\right)^r \cdot \varphi_r(\mathfrak{a}, p\mathfrak{e}). \tag{4}$$

Two rectangles R_1, R_2 are congruent modulo $(\mathfrak{a}, p\mathfrak{e})$ when the rectangles R_1', R_2' formed from their last q columns are congruent modulo p. (The statement that two rectangles of the same degree and the same number of rows are congruent modulo a number means that corresponding elements are congruent modulo that number.) Hence $\varphi_r(\mathfrak{a}, p\mathfrak{e})$ is the number of rectangles, incongruent modulo p, with r rows and q $(\leq r)$ columns, in which not all determinants of degree q are divisible by p. This number, as a simple calculation gives,[3] is

$$p^{qr}\left(1-\frac{1}{p^r}\right)\left(1-\frac{1}{p^{r-1}}\right)\cdots\left(1-\frac{1}{p^{r-q+1}}\right),$$

and therefore, taking (4) into account,

$$\varphi_r(A) = (\mathrm{Nm}\,A)^r\left(1-\frac{1}{p^r}\right)\left(1-\frac{1}{p^{r-1}}\right)\cdots\left(1-\frac{1}{p^{r-q+1}}\right). \tag{5}$$

This equation remains correct also for $q > r$, since in that case the expression on the right side of (5) vanishes.

Thus we have the following theorem:

III. If $A = A_p$ is a primary class and if q of its invariants are divisible by p (the others therefore being equal to 1), then

$$\varphi_r(A) = (\mathrm{Nm}\,A)^r\left(1-\frac{1}{p^r}\right)\left(1-\frac{1}{p^{r-1}}\right)\cdots\left(1-\frac{1}{p^{r-q+1}}\right).$$

It is noteworthy that the expression for $\varphi_r(A)$ does not contain the degree of the class A at all. We now introduce another function into the calculation, in which this degree does play a role: if n is the degree of A, then $\varphi_n(A)$ will be denoted simply by $\varphi(A)$. For the function φ one has the relation

$$\text{IIa.} \qquad \varphi(A) = \prod_p \varphi(A_p).$$

For abbreviation put

$$\vartheta(p,k) = \left(1-\frac{1}{p}\right)\left(1-\frac{1}{p^2}\right)\cdots\left(1-\frac{1}{p^k}\right) \qquad (k>0),$$

$$\vartheta(p,0) = 1.$$

Then Theorem III gives:

IIIa. If $A = A_p$ is a primary class of degree n, and if q of its invariants are divisible by p (so the first $n-q$ are equal to 1), then

$$\varphi(A) = (\mathrm{Nm}\,A)^n \cdot \frac{\vartheta(p,n)}{\vartheta(p,n-q)}.$$

[3]Compare C. Jordan, *Traité des substitutions*, no. 123.

§ 4. Isomorphism; the function χ.

Definition of isomorphism.

Suppose that between the residues ρ of a module $\mathfrak{a}$ and the residues ρ' of a module $\mathfrak{a}'$ one can establish a one-to-one correspondence with the following property: whenever ρ_i, ρ_i' and ρ_k, ρ_k' are two pairs of corresponding residues, then $\rho_i + \rho_k$ and $\rho_i' + \rho_k'$ also form a pair of corresponding residues. Then the correspondence is called an "isomorphic" relation between $\mathfrak{a}$ and $\mathfrak{a}'$. Two modules $\mathfrak{a}, \mathfrak{a}'$ which admit at least one isomorphic relation are called "isomorphic to one another".[4] Thus the relation of isomorphism is always reciprocal.

If $\rho_1, \rho_2, \ldots$ are all residues of $\mathfrak{a}$, and if $\rho_1', \rho_2', \ldots$ are, in this order, the residues of $\mathfrak{a}'$ assigned to them isomorphically, then the isomorphic relation in question will be denoted by

$$\rho_i \parallel \rho_i'.$$

It follows easily from the definition of isomorphism that, in general,

$$\sum_i c_i\rho_i, \qquad \sum_i c_i\rho_i'$$

are corresponding residues of $\mathfrak{a}$ and $\mathfrak{a}'$. If $\mathfrak{a}'$ is isomorphic to $\mathfrak{a}''$ and

$$\rho_i' \parallel \rho_i''$$

denotes an isomorphic relation between $\mathfrak{a}'$ and $\mathfrak{a}''$, then

$$\rho_i \parallel \rho_i''$$

is an isomorphic relation between $\mathfrak{a}$ and $\mathfrak{a}''$; it is called the relation "composed of $\rho_i \parallel \rho_i'$ and $\rho_i' \parallel \rho_i''$". If one composes the isomorphic relation $\rho_i \parallel \rho_i'$ successively with all distinct isomorphic relations from $\mathfrak{a}'$ to $\mathfrak{a}''$, one obtains the same number of isomorphic relations between $\mathfrak{a}$ and $\mathfrak{a}''$, and these exhaust all of them. Thus, if

$$\chi(\mathfrak{a} \parallel \mathfrak{a}')$$

denotes the number of isomorphic relations existing between two modules $\mathfrak{a}$ and $\mathfrak{a}'$, then the following theorem holds:

If $\mathfrak{a}$ is isomorphic to $\mathfrak{a}'$, and $\mathfrak{a}'$ is isomorphic to $\mathfrak{a}''$, then $\mathfrak{a}$ is isomorphic to $\mathfrak{a}''$ and

$$\chi(\mathfrak{a} \parallel \mathfrak{a}'') = \chi(\mathfrak{a}' \parallel \mathfrak{a}'') = \chi(\mathfrak{a} \parallel \mathfrak{a}').$$

The modules $\mathfrak{a}, \mathfrak{a}', \mathfrak{a}''$ need not, of course, all be distinct. Taking $\mathfrak{a} = \mathfrak{a}''$, and assuming the isomorphism of $\mathfrak{a}$ and $\mathfrak{a}'$, one obtains

$$\chi(\mathfrak{a} \parallel \mathfrak{a}') = \chi(\mathfrak{a} \parallel \mathfrak{a}) = \chi(\mathfrak{a}' \parallel \mathfrak{a}').$$

An isomorphic relation of a module $\mathfrak{a}$ onto itself may be interpreted as a permutation among its residues. The permutations so obtained form a group; $\chi(\mathfrak{a} \parallel \mathfrak{a})$, for which we shall also simply write

$$\chi(\mathfrak{a}),$$

is the order of this group.

We now examine more closely under what conditions two modules $\mathfrak{a}$ and $\mathfrak{a}'$ are isomorphic. It is easy to see that this is so when they belong to the same class. For if $[E]$ is a unimodular

[4]The term isomorphism is borrowed from group theory.

transformation carrying $\mathfrak{a}$ into $\mathfrak{a}'$, then (§ 2) it also carries every residue of $\mathfrak{a}$ into a residue of $\mathfrak{a}'$ and in this way establishes a one-to-one correspondence between the residues of $\mathfrak{a}$ and those of $\mathfrak{a}'$, which is immediately recognized as isomorphic.

Conversely, let us now show that two isomorphic modules of the same degree, $\mathfrak{a}, \mathfrak{a}'$, must also belong to the same class. On the basis of the preceding considerations, and without loss of generality, we may assume that $\mathfrak{a}$ and $\mathfrak{a}'$ are the principal modules within their classes. We also exclude the case in which the norms of $\mathfrak{a}$ and $\mathfrak{a}'$ vanish, since it is irrelevant for our purpose. Let

$$\rho_i \parallel \rho_i'$$

be an isomorphic relation between $\mathfrak{a}$ and $\mathfrak{a}'$, let m be any integer, and let ρ_i, ρ_i' run successively through all residues of $\mathfrak{a}, \mathfrak{a}'$. Then the system $m\rho_i$ belongs to the module $\mathfrak{a}$ just as often as the system $m\rho_i'$ belongs to the module $\mathfrak{a}'$. The computation of the number in question is particularly simple when $\mathfrak{a}, \mathfrak{a}'$ are principal modules. If $a_1, a_2, \ldots, a_n$ are the invariants of $\mathfrak{a}$, and $a_1', a_2', \ldots, a_n'$ those of $\mathfrak{a}'$, one immediately sees that the number of residues ρ_i for which $m\rho_i$ is contained in $\mathfrak{a}$ is

$$(m, a_1)(m, a_2)\cdots(m, a_n),$$

and similarly, the number of residues ρ_i' for which $m\rho_i'$ belongs to $\mathfrak{a}'$ is

$$(m, a_1')(m, a_2')\cdots(m, a_n').$$

Hence for every integer m the equality

$$(m, a_1)(m, a_2)\cdots(m, a_n) = (m, a_1')(m, a_2')\cdots(m, a_n')$$

must hold; and this, as a simple consideration shows, can occur only when

$$a_1 = a_1', \quad a_2 = a_2', \quad \ldots, \quad a_n = a_n',$$

that is, when $\mathfrak{a}$ and $\mathfrak{a}'$ belong to the same class.

Nevertheless two modules of different degrees may certainly be isomorphic; the question of the isomorphism of two modules is settled by the following theorem.

I. Two modules are isomorphic if and only if their systems of invariants are identical after the invariants which reduce to 1 have been removed.

To prove that the stated condition is sufficient, suppose that the module $\mathfrak{a}$ has invariants

$$a_1 = 1, \ldots, a_r = 1, a_{r+1}, a_{r+2}, \ldots, a_n,$$

and that the module $\mathfrak{a}'$ has invariants

$$a_1' = a_{r+1}, \quad a_2' = a_{r+2}, \quad \ldots, \quad a_{n-r}' = a_n.$$

We must show that $\mathfrak{a}$ and $\mathfrak{a}'$ are isomorphic. We may assume $\mathfrak{a}$ and $\mathfrak{a}'$ to be principal modules, and under this assumption the proof is very simple. Let

$$\rho_1, \rho_2, \ldots$$

be a complete system of residues with respect to $\mathfrak{a}$, and let ρ_i' denote the row obtained from ρ_i by deleting the first r elements. Then it follows at once that

$$\rho_1', \rho_2', \ldots$$

form a complete system of residues of $\mathfrak{a}'$, and that

$$\rho_i \parallel \rho_i'$$

defines an isomorphic relation between $\mathfrak{a}$ and $\mathfrak{a}'$. Combining this result with the earlier one, according to which modules of the same degree are isomorphic only when they belong to the same class, makes clear the necessity of the condition stated in Theorem I.

Some simple consequences for the function χ may be drawn from this theorem. First, one sees that for all modules $\mathfrak{a}$ of a class A, χ has the same value, so that the symbol $\chi(\mathfrak{a})$ may conveniently be replaced by

$$\chi(A).$$

Moreover one has:

II. If the systems of invariants of the classes A and A' are identical after the invariants reducing to 1 have been removed, then

$$\chi(A) = \chi(A').$$

§ 5. Relation among the functions φ, χ, ψ.

The functions φ, χ, ψ considered here stand in a very simple relation to one another, made clear by the following argument.

Let A be a class of degree n, let $\mathfrak{a}$ be a module in A, and let R be an r-rowed residue of $\mathfrak{a}$. Then, as a simple consideration shows, the following theorem holds: in order that

$$(\mathfrak{a}, R) = \mathfrak{e} \tag{1}$$

hold, it is necessary and sufficient that for every row ρ of n elements one can find a row σ of r elements satisfying the congruence

$$\sigma R \equiv \rho \pmod{\mathfrak{a}}. \tag{2}$$

Now suppose that (1) is satisfied and that $r = n$. Then all σ for which σR is contained in $\mathfrak{a}$ form a module $\mathfrak{a}'$ of degree n; all σ satisfying a congruence of the form (2), where ρ is regarded as given, are representatives of a definite residue of $\mathfrak{a}'$; and conversely, for all representatives of a definite residue of $\mathfrak{a}'$, σR represents one and the same residue of $\mathfrak{a}$. Thus one obtains a one-to-one correspondence between the residues of $\mathfrak{a}$ and those of $\mathfrak{a}'$ by making the residue σ of $\mathfrak{a}'$ correspond to the residue σR of $\mathfrak{a}$. This correspondence is, as one easily sees, an isomorphism. It follows that the module $\mathfrak{a}'$ belongs to the class A (§ 4, I). We shall say that the residue R of $\mathfrak{a}$ belongs to the module $\mathfrak{a}'$, and we shall denote the isomorphic relation mediated by R by $[R]$.

If R' is also an n-rowed residue of $\mathfrak{a}$ satisfying condition (1), but distinct from R, then R' may well belong to the same module $\mathfrak{a}'$, but the isomorphic relation $[R']$ is different from $[R]$. Otherwise, for every row σ the congruence

$$\sigma R \equiv \sigma R' \pmod{\mathfrak{a}}$$

would have to hold; and by putting for σ successively the rows of the system

$$E^0 = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix},$$

one would infer the congruence

$$R' \equiv R \pmod{\mathfrak{a}}.$$

On the other hand, it is easy to show that, when R is allowed to run through every n-rowed residue of $\mathfrak{a}$ satisfying (1), the symbol $[R]$ represents exactly once every isomorphic relation possible between $\mathfrak{a}$ and any module $\mathfrak{a}'$ of A. Indeed, suppose such a relation between $\mathfrak{a}$ and $\mathfrak{a}'$ is given. If it assigns to the residues of $\mathfrak{a}'$ represented by the n rows of E^0 the residues $\rho_1, \rho_2, \ldots, \rho_n$ of $\mathfrak{a}$, and if R is the n-rowed residue formed from them, then the definition of isomorphism gives that the residue σ of $\mathfrak{a}'$ must correspond to the residue σR of $\mathfrak{a}$. Since every residue of $\mathfrak{a}$ is thus representable in the form σR, it first follows that R satisfies

$$(\mathfrak{a}, R) = \mathfrak{e},$$

and then that the given isomorphic relation is identical with the one denoted by $[R]$.

The result of these considerations is that each of the residues R, whose number is $\varphi(A)$, belongs to a definite one of the $\psi(A)$ modules of the class A, and that to each module $\mathfrak{a}'$ of A there belong $\chi(A)$ residues R, namely as many as there are isomorphic relations between $\mathfrak{a}$ and $\mathfrak{a}'$. Hence one obtains the important relation

$$\text{I.} \qquad \varphi(A) = \chi(A) \cdot \psi(A).$$

From this and the formulas (§ 3, II; § 2, XVII) one obtains further

$$\chi(A) = \prod_p \chi(A_p),$$

which of course could also be proved directly.

§ 6. Determination of the functions ψ and χ.

From the theorems § 2, IX; § 4, II; § 5, I it is now easy to express the functions ψ and χ, first of all, as quotients of products of φ-functions.

Let $A = \mathrm{Cl}(a_1 \mid a_2 \mid \cdots \mid a_n)$ be any class of degree n and positive norm, and put

$$A' = \mathrm{Cl}\left(1 \mid \frac{a_2}{a_1} \mid \frac{a_3}{a_1} \mid \cdots \mid \frac{a_n}{a_1}\right),$$

$$A_i = \mathrm{Cl}\left(\frac{a_{i+1}}{a_i} \mid \frac{a_{i+2}}{a_i} \mid \cdots \mid \frac{a_n}{a_i}\right), \qquad A'_i = \mathrm{Cl}\left(1 \mid \frac{a_{i+2}}{a_{i+1}} \mid \frac{a_{i+3}}{a_{i+1}} \mid \cdots \mid \frac{a_n}{a_{i+1}}\right),$$

so that A_i, A'_i are classes of degree $n-i$. Then

$$A = a_1 A', \qquad A_i = \frac{a_{i+1}}{a_i} A'_i,$$

$$\psi(A) = \psi(A'), \quad \psi(A_i) = \psi(A'_i) \quad (\S 2,\ IX),$$

$$\psi(A') = \frac{\varphi(A')}{\chi(A')}, \quad \psi(A'_i) = \frac{\varphi(A'_i)}{\chi(A'_i)} \quad (\S 5,\ I),$$

$$\chi(A') = \chi(A_1), \quad \chi(A'_i) = \chi(A_{i+1}) \quad (\S 4,\ II),$$

$$\chi(A_1) = \frac{\varphi(A_1)}{\psi(A_1)}, \quad \chi(A_{i+1}) = \frac{\varphi(A_{i+1})}{\psi(A_{i+1})} \quad (\S 5,\ I),$$

and hence

$$\psi(A) = \frac{\varphi(A')}{\varphi(A_1)} \psi(A_1), \qquad \psi(A_i) = \frac{\varphi(A'_i)}{\varphi(A_{i+1})} \psi(A_{i+1}).$$

Since $\psi(A_{n-1}) = 1$, these relations give

$$\text{I.}\qquad \psi(A) = \frac{\varphi(A')\varphi(A'_1)\cdots\varphi(A'_{n-2})}{\varphi(A_1)\varphi(A_2)\cdots\varphi(A_{n-1})},$$

$$\text{II.}\qquad \chi(A) = \frac{\varphi(A)\varphi(A_1)\cdots\varphi(A_{n-1})}{\varphi(A')\varphi(A'_1)\cdots\varphi(A'_{n-2})}.$$

The further calculation, which we shall carry out only for the function ψ, first requires treating the case of a primary class. Thus let

$$\mathrm{Cl}(a_1 \mid a_2 \mid \cdots \mid a_n) = A = A_p = \mathrm{Cl}(p^{r_1} \mid p^{r_2} \mid \cdots \mid p^{r_n})$$

and consequently

$$r_1 \leq r_2 \leq \cdots \leq r_n.$$

Let $e_1, e_2, \ldots, e_k$ denote the distinct exponents among the r's, in their natural order. If e_1 occurs s_1 times, e_2 occurs s_2 times, and so on, e_k occurs s_k times among the exponents r, then

$$s_1 + s_2 + \cdots + s_k = n. \tag{1}$$

Among the differences $r_m - r_l$ with $m > l$, the difference $e_i - e_h$ $(i > h)$ occurs $s_i s_h$ times. Hence

$$\sum_{i>h}(e_i - e_h)s_i s_h = \sum_{m>l} r_m - r_l = (1-n)r_1 + (3-n)r_2 + (5-n)r_3 + \cdots + (n-3)r_{n-1} + (n-1)r_n. \tag{2}$$

Now put

$$s_1 + s_2 + \cdots + s_i = t_i \qquad (i = 1, 2, \ldots, k).$$

From

$$p^{e_1} = a_1 = a_2 = \cdots = a_{t_1}, \quad p^{e_2} = a_{t_1+1} = \cdots = a_{t_2}, \quad \ldots, \quad p^{e_k} = a_{t_{k-1}+1} = \cdots = a_{t_k}, \tag{3}$$

it follows that in (I) only the classes

$$A',\ A'_1, \ldots, A'_{k-2} \quad \text{and} \quad A_1, \ldots, A_{k-1}$$

occur essentially distinctly. More precisely,

$$\psi(A) = \frac{\varphi(A')\varphi(A'_1)\cdots\varphi(A'_{k-2})}{\varphi(A_1)\varphi(A_2)\cdots\varphi(A_{k-1})}. \tag{4}$$

The classes A', A_i, A'_i have degrees n, $n - t_i$, $n - t_i$, respectively, while the numbers of their invariants divisible by p are respectively $n - t_1$, $n - t_i$, $n - t_{i+1}$. Therefore (§ 3, IIIa)

$$\varphi(A') = (\mathrm{Nm}\, A')^n \frac{\vartheta(p; n)}{\vartheta(p; t_1)} = (\mathrm{Nm}\, A')^n \frac{\vartheta(p; n)}{\vartheta(p; s_1)},$$

$$\varphi(A_i) = (\mathrm{Nm}\, A_i)^{n-t_i}\vartheta(p; n - t_i),$$

$$\varphi(A'_i) = (\mathrm{Nm}\, A'_i)^{n-t_i} \frac{\vartheta(p; n - t_i)}{\vartheta(p; t_{i+1} - t_i)} = (\mathrm{Nm}\, A'_i)^{n-t_i} \frac{\vartheta(p; n - t_i)}{\vartheta(p; s_{i+1})}.$$

Thus, by (4),

$$\psi(A) = \frac{(\mathrm{Nm}\, A')^n (\mathrm{Nm}\, A'_1)^{n-t_1} \cdots (\mathrm{Nm}\, A'_{k-2})^{n-t_{k-2}}}{(\mathrm{Nm}\, A_1)^{n-t_1}(\mathrm{Nm}\, A_2)^{n-t_2} \cdots (\mathrm{Nm}\, A_{k-1})^{n-t_{k-1}}} \frac{\vartheta(p; n)}{\vartheta(p; s_1)\vartheta(p; s_2)\cdots\vartheta(p; s_k)}. \tag{5}$$

Now

$$\frac{(\mathrm{Nm}\,A'_{i-1})^{n-t_{i-1}}}{(\mathrm{Nm}\,A_i)^{n-t_i}} = \left(\frac{\mathrm{Nm}\,A'_{i-1}}{\mathrm{Nm}\,A'_i}\right)^{n-t_i} (\mathrm{Nm}\,A'_{i-1})^{t_i-t_{i-1}},$$

and hence

$$\frac{(\mathrm{Nm}\,A')^n(\mathrm{Nm}\,A'_1)^{n-t_1}\cdots(\mathrm{Nm}\,A'_{k-2})^{n-t_{k-2}}}{(\mathrm{Nm}\,A_1)^{n-t_1}(\mathrm{Nm}\,A_2)^{n-t_2}\cdots(\mathrm{Nm}\,A_{k-1})^{n-t_{k-1}}} = \left(\frac{\mathrm{Nm}\,A'}{\mathrm{Nm}\,A_1}\right)^{t_1}\left(\frac{\mathrm{Nm}\,A'_1}{\mathrm{Nm}\,A_2}\right)^{t_2}\cdots\left(\frac{\mathrm{Nm}\,A'_{k-3}}{\mathrm{Nm}\,A_{k-2}}\right)^{t_{k-2}}(\mathrm{Nm}\,A'_{k-2})^{t_{k-1}}. \tag{6}$$

Furthermore,

$$\frac{\mathrm{Nm}\,A'_{i-1}}{\mathrm{Nm}\,A'_i} = \left(\frac{a_{t_i+1}}{a_{t_i}}\right)^{n-t_i}, \qquad \mathrm{Nm}\,A'_{k-2} = \left(\frac{a_n}{a_{t_{k-1}}}\right)^{n-t_{k-1}},$$

$$\prod_{i=1}^{k-2}\left(\frac{\mathrm{Nm}\,A'_{i-1}}{\mathrm{Nm}\,A'_i}\right)^{t_i}(\mathrm{Nm}\,A'_{k-2})^{t_{k-1}} = \prod_{i=1}^{k-1}\left(\frac{a_{t_i+1}}{a_{t_i}}\right)^{(n-t_i)t_i} = \prod_{i=1}^{n-1}\left(\frac{a_{i+1}}{a_i}\right)^{i(n-i)} = \prod_{i=1}^{n} a_i^{2i-1-n}. \tag{7}$$

From (5), (6), (7) one now obtains

$$\text{III.} \qquad \psi(A) = a_1^{1-n}a_2^{3-n}a_3^{5-n}\cdots a_{n-1}^{n-3}a_n^{n-1}\frac{\vartheta(p;n)}{\vartheta(p;s_1)\vartheta(p;s_2)\cdots\vartheta(p;s_k)}.$$

or, taking (2), (3) into account,

$$\text{IIIa.} \qquad \psi(A) = p^{\sum_{i>h} s_i s_h (e_i-e_h)}\frac{\vartheta(p;n)}{\vartheta(p;s_1)\vartheta(p;s_2)\cdots\vartheta(p;s_k)}.$$

Introduce a function $g(x;n,m)$[5] by the equation

$$g(x;n,m) = \frac{(1-x^n)(1-x^{n-1})\cdots(1-x^{n-m+1})}{(1-x)(1-x^2)\cdots(1-x^m)}, \tag{8}$$

where n is an integer and m a positive integer. Then

$$g(x;n,m) = g(x;n-1,m) + x^{n-m}g(x;n-1,m-1), \tag{9}$$

and this equation also holds for arbitrary integral m if one stipulates that

$$g(x;n,0) = 1, \tag{10}$$

$$g(x;n,m) = 0 \qquad (m<0). \tag{11}$$

There is also the relation

$$g(x;n,m) = x^{m(n-m)}g\left(\frac{1}{x};n,m\right). \tag{12}$$

From (9), (10), (11) it follows that $g(x;n,m)$ is, for $n \geq 0$, an integral function of x with only nonnegative integer coefficients. For $n \geq m \geq 0$ the functions g and ϑ satisfy

$$g\left(\frac{1}{x};n,m\right) = \frac{\vartheta(x;n)}{\vartheta(x;m)\vartheta(x;n-m)}. \tag{13}$$

Writing, for abbreviation,

$$\psi(A) = p^{\sum_{i>h} s_i s_h (e_i-e_h)}\bar{\psi}(A) = a_1^{1-n}a_2^{3-n}\cdots a_n^{n-1}\bar{\psi}(A), \tag{14}$$

[5]It is the same function denoted by (n,m) in Gauss's paper "Summatio quarundam serierum singularium".

one obtains the following expressions for the functions ψ, $\bar{\psi}$:

$$\bar{\psi}(A) = g\left(\frac{1}{p}; n, s_1\right) g\left(\frac{1}{p}; n - s_1, s_2\right) \cdots g\left(\frac{1}{p}; n - s_1 - \cdots - s_{k-2}, s_{k-1}\right), \tag{15}$$

IIIb. $$\psi(A) = p^{\sum_{i>h} s_i s_h (e_i - e_h - 1)} g(p; n, s_1) g(p; n - s_1, s_2) \cdots g(p; n - s_1 - \cdots - s_{k-2}, s_{k-1}). \tag{16}$$

This representation shows that $\psi(A)$ is an integral function of p, that the coefficient of the highest power of p is equal to 1, and that the remaining coefficients are nonnegative integers.

For an arbitrary class A of positive norm, § 2, XVII reduces the computation of $\psi(A)$ to the case of a primary class. If one sets

$$\prod_p \bar{\psi}(A_p) = \bar{\psi}(A),$$

then

$$\psi(A) = \prod_p \Big((a_1)_p^{1-n} (a_2)_p^{3-n} \cdots (a_n)_p^{n-1} \bar{\psi}(A_p) \Big) = a_1^{1-n} a_2^{3-n} \cdots a_n^{n-1} \bar{\psi}(A),$$

so that equation (14) holds in general.

Applying the results obtained to the case $n = 2$, one obtains, for $A = \mathrm{Cl}(a_1 \mid a_2)$,

$$\psi(A) = \frac{a_2}{a_1} \bar{\psi}(A) = \frac{a_2}{a_1} \prod_{p \mid a_2/a_1} \bar{\psi}(A_p) = \frac{a_2}{a_1} \prod_{p \mid a_2/a_1} \frac{\vartheta(p;2)}{(\vartheta(p;1))^2} = \frac{a_2}{a_1} \prod_{p \mid a_2/a_1} \left(1 + \frac{1}{p}\right),$$

where the product extends over the primes dividing a_2/a_1.

Example. For

$$a_1 = 3, \qquad a_2 = 90$$

one has

$$\frac{a_2}{a_1} = 30 = 2 \cdot 3 \cdot 5,$$

$$\psi(A) = 30 \cdot \frac{3}{2} \cdot \frac{4}{3} \cdot \frac{6}{5} = 72.$$

For

$$n = 3, \qquad A = \mathrm{Cl}(a_1 \mid a_2 \mid a_3),$$

one has

$$\psi(A) = \left(\frac{a_3}{a_1}\right)^2 \bar{\psi}(A) = \left(\frac{a_3}{a_1}\right)^2 \prod_{p \mid a_3/a_1} \frac{\vartheta(p;3)}{\vartheta(p;1)\vartheta(p;2)} \prod_{\substack{p \mid a_2/a_1 \\ p \mid a_3/a_2}} \frac{\vartheta(p;2)}{\vartheta(p;1)\vartheta(p;1)},$$

so

$$\psi(A) = \left(\frac{a_3}{a_1}\right)^2 \prod_{p \mid a_3/a_1} \left(1 + \frac{1}{p} + \frac{1}{p^2}\right) \prod_{\substack{p \mid a_2/a_1 \\ p \mid a_3/a_2}} \left(1 + \frac{1}{p}\right),$$

where the first product extends over the primes dividing a_3/a_1, and the second over the primes that divide both a_2/a_1 and a_3/a_2.

Example. For

$$a_1 = 2, \qquad a_2 = 210, \qquad a_3 = 6930$$

one has

$$\frac{a_3}{a_1} = 3465 = 3^2 \cdot 5 \cdot 7 \cdot 11, \qquad \frac{a_2}{a_1} = 105 = 3 \cdot 5 \cdot 7, \qquad \frac{a_3}{a_2} = 33 = 3 \cdot 11.$$

$$\psi(A) = (3465)^2 \cdot \frac{13}{9} \cdot \frac{31}{25} \cdot \frac{57}{49} \cdot \frac{133}{121} \cdot \frac{4}{3} = 36661716.$$

§ 7. Introduction of several functions depending on several classes.

The function ψ forms the basis for determining other functions which are important for the theory of modules and which will be treated below. We shall consider only modules and classes of positive norm.

A module $\mathfrak{a}$ of positive norm has, as one easily sees, only finitely many divisors, which (§ 2, III) are distributed among those classes that are divisors of $\mathrm{Cl}\,\mathfrak{a}$. We now focus on the divisors of $\mathfrak{a}$ which belong to a specified class B. If

$$\mathfrak{b}_1, \mathfrak{b}_2, \ldots, \mathfrak{b}_k$$

are these modules and E denotes a unimodular system, then (§ 2, VI)

$$\mathfrak{b}_1 E, \mathfrak{b}_2 E, \ldots, \mathfrak{b}_k E$$

are all divisors of $\mathfrak{a}E$ within the class B. It follows that the number of divisors of $\mathfrak{a}$ within B depends only on the classes $A = \mathrm{Cl}\,\mathfrak{a}$ and B. We denote this number by

$$t(A \mid B).$$

Similarly, the number of multiples which a module $\mathfrak{b}$ of class B possesses within the class A is a function solely of the classes A and B. Let it be denoted by

$$v(A \mid B).$$

The number of pairs of modules $\mathfrak{a}_i, \mathfrak{b}_k$ satisfying

$$\mathrm{Cl}\,\mathfrak{a}_i = A, \qquad \mathrm{Cl}\,\mathfrak{b}_k = B, \qquad (\mathfrak{a}_i, \mathfrak{b}_k) = \mathfrak{b}_k$$

is represented by each of the expressions $\psi(A)t(A \mid B)$ and $\psi(B)v(A \mid B)$. Thus

$$\text{I.} \qquad \psi(A)t(A \mid B) = \psi(B)v(A \mid B).$$

Furthermore § 2, XV gives the formulas

$$\text{II.} \qquad t(A \mid B) = \prod_p t(A_p \mid B_p),$$

$$\text{III.} \qquad v(A \mid B) = \prod_p v(A_p \mid B_p).$$

From § 2, III we know that $t(A \mid B)$ and $v(A \mid B)$ are nonzero or zero according as A is or is not divisible by B. Finally, for every class A one has (§ 2, IV)

$$\text{IV.} \qquad t(A \mid A) = v(A \mid A) = 1.$$

Other class-functions are reached by the following considerations.

If the modules $\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s$, which need not all be distinct, are divisors of a module $\mathfrak{c}$, then their least common multiple $[\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s]$ is also a divisor of $\mathfrak{c}$. One may therefore pose the problem of determining, for a given $\mathfrak{c}$, s modules $\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s$ such that

$$[\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_s] = \mathfrak{c}.$$

This problem is of course always solvable, and in general in many ways. It is different, however, if one imposes the condition that each module $\mathfrak{a}_i$ shall belong to a specified given class A_i.

From § 2, VII follows the theorem: if for some module $\mathfrak{c}$ of class Γ it is possible to represent it as the least common multiple of s modules $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s$ belonging respectively to the classes $A_1, A_2, \dots, A_s$, then the same is possible for every module of the class Γ. More generally, the number of systems of s modules $\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s$ satisfying

$$\mathrm{Cl}\,\mathfrak{a}_1 = A_1, \quad \mathrm{Cl}\,\mathfrak{a}_2 = A_2, \quad \dots \quad \mathrm{Cl}\,\mathfrak{a}_s = A_s; \qquad [\mathfrak{a}_1, \mathfrak{a}_2, \dots, \mathfrak{a}_s] = \mathfrak{c}$$

is the same for every module $\mathfrak{c}$ of class Γ. We denote this number by

$$\bar{\gamma}(\Gamma; A_1, A_2, \dots, A_s).$$

It is a function of the classes $\Gamma, A_1, \dots, A_s$, symmetric in $A_1, \dots, A_s$. The question whether a module $\mathfrak{c}$ of class Γ can be represented as the least common multiple of s modules $\mathfrak{a}_1, \dots, \mathfrak{a}_s$ belonging respectively to the classes $A_1, \dots, A_s$ is therefore exactly the question whether $\bar{\gamma}(\Gamma; A_1, \dots, A_s) > 0$. In any given case this can of course be decided by a finite number of trials. The problem, however, is whether one can state general, readily surveyable relations whose existence among the classes $\Gamma, A_1, \dots, A_s$ is the necessary and sufficient condition for the nonvanishing of the function $\bar{\gamma}(\Gamma; A_1, \dots, A_s)$.

First one sees that $\bar{\gamma}(\Gamma; A_1, \dots, A_s)$ can be nonzero only when Γ is divisible by each of the classes $A_1, \dots, A_s$. Thus, if the classes $A_1, \dots, A_s$ are regarded as given, their least common multiple $\Gamma' = [A_1, \dots, A_s]$ is, in a sense, the minimum among the classes Γ for which $\bar{\gamma}(\Gamma; A_1, \dots, A_s) > 0$: first, $\bar{\gamma}(\Gamma'; A_1, \dots, A_s) > 0$, since the principal module of Γ' is the least common multiple of the principal modules of $A_1, \dots, A_s$; and second, every class Γ satisfying $\bar{\gamma}(\Gamma; A_1, \dots, A_s) > 0$ is a multiple of Γ'. A further investigation shows that among the classes Γ under discussion there also exists a maximum, namely one of them, call it Γ'', that is divisible by all the others. The construction of the class Γ'' will be given shortly.

Entirely analogous results are obtained in the investigation of the greatest common divisor of modules. We denote by

$$\bar{\delta}(\Delta; A_1, \dots, A_s)$$

the number of systems of s modules $\mathfrak{a}_1, \dots, \mathfrak{a}_s$ respectively taken from the classes $A_1, \dots, A_s$ and having a given module $\mathfrak{b}$ of class Δ as their greatest common divisor. If the classes $A_1, \dots, A_s$ are regarded as given, then all classes Δ for which $\bar{\delta}(\Delta; A_1, \dots, A_s) > 0$ lie between a maximum Δ' and a minimum Δ'', so that

$$\bar{\delta}(\Delta'; A_1, \dots, A_s) > 0, \qquad \bar{\delta}(\Delta''; A_1, \dots, A_s) > 0,$$

and every other class of the kind considered is a divisor of Δ' and a multiple of Δ''. Here Δ' is plainly the greatest common divisor of the classes $A_1, \dots, A_s$.

To obtain the classes Γ'', Δ'', proceed as follows. Let

$$A_i = \mathrm{Cl}(a_{i1} \mid a_{i2} \mid \cdots \mid a_{in}) \qquad (i = 1, 2, \dots, s).$$

From all $s \cdot n$ invariants

$$a_{11}, \dots, a_{1n}, \dots, a_{s1}, \dots, a_{sn}$$

of the classes $A_1, \dots, A_s$, form a diagonal system K and determine its class K. This has degree $s \cdot n$ and is plainly independent of the order of the classes $A_1, \dots, A_s$. The first n invariants of K form the invariant system of a class of degree n, which we call the "limiting divisor" of the classes $A_1, \dots, A_s$ and denote by

$$(\!| A_1, \dots, A_s |\!).$$

Similarly, the last n invariants of K form the invariant system of a class of degree n. This class is called the "limiting multiple" of the classes $A_1, \dots, A_s$ and is denoted by

$$[\![A_1, \dots, A_s]\!].$$

It is the maximum mentioned above for the classes Γ for which $\bar{\gamma}(\Gamma; A_1, \ldots, A_s) > 0$; likewise, the limiting divisor of $A_1, \ldots, A_s$ is the minimum of the classes Δ for which $\bar{\delta}(\Delta; A_1, \ldots, A_s) > 0$. It is not hard to prove this without entering into a general determination of the functions $\bar{\gamma}$ and $\bar{\delta}$. But the question now arises whether conversely, for all classes Γ lying between $[\![A_1, \ldots, A_s]\!]$ and $[A_1, \ldots, A_s]$, and for all classes Δ lying between $(A_1, \ldots, A_s)$ and $(\!|A_1, \ldots, A_s|\!)$, the functions $\bar{\gamma}(\Gamma; A_1, \ldots, A_s)$ and $\bar{\delta}(\Delta; A_1, \ldots, A_s)$ are positive. I was able to decide this question only by a detailed study of the functions $\bar{\gamma}$ and $\bar{\delta}$; it turned out that the question just raised must be answered affirmatively. Thus one obtains the following double theorem, which is also of interest especially for the theory of groups with commuting elements:

V. a. In order that a module $\mathfrak{c}$ of class Γ can be represented as the least common multiple of s modules $\mathfrak{a}_1, \ldots, \mathfrak{a}_s$ belonging respectively to the classes $A_1, \ldots, A_s$, it is necessary and sufficient that Γ lie between the limiting multiple and the least common multiple of the classes $A_1, \ldots, A_s$.

b. In order that a module $\mathfrak{b}$ of class Δ can be represented as the greatest common divisor of s modules $\mathfrak{a}_1, \ldots, \mathfrak{a}_s$ belonging respectively to the classes $A_1, \ldots, A_s$, it is necessary and sufficient that Δ lie between the greatest common divisor and the limiting divisor of the classes $A_1, \ldots, A_s$.

In what follows, the proof of this theorem is completely supplied by the investigation of the functions $\bar{\gamma}$ and $\bar{\delta}$.

From § 2, XVI follow the relations

$$\text{VI.} \qquad \bar{\gamma}(\Gamma; A_1, \ldots, A_s) = \prod_p \bar{\gamma}(\Gamma_p; (A_1)_p, \ldots, (A_s)_p),$$

$$\bar{\delta}(\Delta; A_1, \ldots, A_s) = \prod_p \bar{\delta}(\Delta_p; (A_1)_p, \ldots, (A_s)_p).$$

It is also easy to see from this that, in order to prove V, it is necessary only to prove it for primary classes.

§ 8. The functions t and v expressed by the function ψ.

We now turn to a closer examination of the functions $t(A \mid B)$ and $v(A \mid B)$, and first show how they can be represented with the help of ψ-functions. This can be done in various ways.

Let $A = \mathrm{Cl}(a_1 \mid a_2 \mid \cdots \mid a_n)$, $B = \mathrm{Cl}(b_1 \mid b_2 \mid \cdots \mid b_n)$, let $\mathfrak{a}$ be an arbitrary module of A, and finally let

$$\mathfrak{b}_1, \mathfrak{b}_2, \ldots, \mathfrak{b}_{\psi(B)} \tag{1}$$

be all modules of B. The module $\mathfrak{a}$ is divisible by $a_1\mathfrak{e}$ (§ 2, X); hence, if $\mathfrak{a}$ is divisible by $\mathfrak{b}_k$, it is also divisible by $[a_1\mathfrak{e}, \mathfrak{b}_k]$, and conversely. The modules

$$[a_1\mathfrak{e}, \mathfrak{b}_1], [a_1\mathfrak{e}, \mathfrak{b}_2], \ldots, [a_1\mathfrak{e}, \mathfrak{b}_{\psi(B)}] \tag{2}$$

all belong to the class $[a_1E, B]$ (§ 2, XI), and a simple consideration shows that every module of $[a_1E, B]$ occurs in the sequence (2) equally often, namely

$$\frac{\psi(B)}{\psi([a_1E, B])}$$

times. Since $\mathfrak{a}$ is divisible by $t(A \mid [a_1E, B])$ modules of the class $[a_1E, B]$, one obtains:

I. If a_1 is the first invariant of the class A, then

$$t(A \mid B) = \frac{\psi(B)}{\psi([a_1E, B])} \cdot t(A \mid [a_1E, B]).$$

Furthermore:

II. If the classes

$$A = \mathrm{Cl}(a_1 \mid a_2 \mid \cdots \mid a_n) \quad \text{and} \quad B = \mathrm{Cl}(b_1 \mid b_2 \mid \cdots \mid b_n)$$

have the same first invariant $a_1 = b_1$, and if

$$A' = \mathrm{Cl}(a_2 \mid \cdots \mid a_n), \qquad B' = \mathrm{Cl}(b_2 \mid \cdots \mid b_n)$$

is put, then

$$t(A \mid B) = t(A' \mid B').$$

Proof. Let A be the principal system of A, let α be its first row, let $\mathrm{Md}\, A = \mathfrak{a}$, let $\mathfrak{a}'$ be the principal module of A', and let $\mathfrak{b} = \mathrm{Md}\, B$ be any divisor of $\mathfrak{a}$ within the class B. If

$$B = \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & & \vdots \\ b_{n1} & \cdots & b_{nn} \end{pmatrix}$$

and if $\beta_1, \beta_2, \ldots, \beta_n$ denote the rows of B, then first $\mathfrak{b} = \mathrm{Md}(\beta_1, \ldots, \beta_n)$; and since $\mathfrak{a}$ is divisible by $\mathfrak{b}$, so that the row α belongs to the module $\mathfrak{b}$, one has

$$\mathfrak{b} = \mathrm{Md}(\alpha, \beta_1, \ldots, \beta_n).$$

From $\mathrm{Cl}\, B = B$ and $a_1 = b_1$ it follows that all elements b_{i1} are divisible by a_1. Put $b_{i1} = m_i a_1$ $(i = 1, \ldots, n)$ and $\beta_i - m_i \alpha = \gamma_i$. Then

$$\mathfrak{b} = \mathrm{Md}(\alpha, \gamma_1, \ldots, \gamma_n).$$

In γ_i the first element is 0; let γ_i' denote the row of $n-1$ elements left after deleting this zero. Then $\mathfrak{b}' = \mathrm{Md}(\gamma_1', \ldots, \gamma_n')$ is a module of degree $n-1$. If

$$B' = \begin{pmatrix} b'_{11} & \cdots & b'_{1,n-1} \\ \vdots & & \vdots \\ b'_{n-1,1} & \cdots & b'_{n-1,n-1} \end{pmatrix} \tag{3}$$

is any square basis of this module, then

$$\bar{B} = \begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & b'_{11} & \cdots & b'_{1,n-1} \\ \vdots & \vdots & & \vdots \\ 0 & b'_{n-1,1} & \cdots & b'_{n-1,n-1} \end{pmatrix} \tag{4}$$

is a basis of $\mathfrak{b}$. From $\mathrm{Cl}\, \bar{B} = B$ it follows easily that $\mathrm{Cl}\, \mathfrak{b}' = \mathrm{Cl}\, B' = B'$, and from the divisibility of $\mathfrak{a}$ by $\mathfrak{b}$ it follows that $\mathfrak{a}'$ is divisible by $\mathfrak{b}'$. Conversely, the following is easy to see: if one has two square systems B', $\bar{B}$ represented as in (3) and (4), and if $\mathrm{Md}\, B' = \mathfrak{b}'$ is a divisor of $\mathfrak{a}'$ with $\mathrm{Cl}\, B' = B'$, then $\mathrm{Md}\, \bar{B} = \mathfrak{b}$ is a divisor of $\mathfrak{a}$ and $\mathrm{Cl}\, \bar{B} = B$. Finally, one sees that, in the way described, the module $\mathfrak{b}'$ is uniquely determined by $\mathfrak{b}$, and likewise $\mathfrak{b}$ by $\mathfrak{b}'$. Thus $\mathfrak{a}$ has just as many divisors in the class B as $\mathfrak{a}'$ has in the class B', that is,

$$t(A \mid B) = t(A' \mid B').$$

Now let $A = \mathrm{Cl}(a_1 \mid a_2 \mid \cdots \mid a_n)$ be divisible by $B = \mathrm{Cl}(b_1 \mid b_2 \mid \cdots \mid b_n)$, and put

$$\begin{cases} \mathrm{Cl}(a_{i+1} \mid a_{i+2} \mid \cdots \mid a_n) = A_i, \\ \mathrm{Cl}([a_i, b_{i+1}] \mid [a_i, b_{i+2}] \mid \cdots \mid [a_i, b_n]) = B_i, \quad (i = 0, 1, \ldots, n-1). \\ \mathrm{Cl}([a_{i+1}, b_{i+1}] \mid [a_{i+1}, b_{i+2}] \mid \cdots \mid [a_{i+1}, b_n]) = \bar{B}_i, \end{cases} \tag{5}$$

Then

$$\bar{B}_i = [a_{i+1}E, B_i]$$

and hence by Theorem I

$$t(A_i \mid B_i) = \frac{\psi(B_i)}{\psi(\bar{B}_i)}\, t(A_i \mid \bar{B}_i).$$

Moreover $[a_{i+1}, b_{i+1}] = a_{i+1}$, whence by Theorem II

$$t(A_i \mid \bar{B}_i) = t(A_{i+1} \mid B_{i+1}).$$

Finally, taking into account that $t(A_{n-1} \mid B_{n-1}) = 1$, one obtains the formula

$$\text{III.} \qquad t(A \mid B) = \frac{\psi(B)\psi(B_1)\cdots\psi(B_{n-2})}{\psi(\bar{B})\psi(\bar{B}_1)\cdots\psi(\bar{B}_{n-2})}.$$

The function $v(A \mid B)$ can also be represented as a quotient of products of ψ-functions by using two theorems that form a dual analogue of I and II. The same result can also be obtained by applying the formulas § 7, I and the formula just found for $t(A \mid B)$. One then obtains

$$\text{IV.} \qquad v(A \mid B) = \frac{\psi(A)\psi(B_1)\cdots\psi(B_{n-2})}{\psi(\bar{B})\psi(\bar{B}_1)\cdots\psi(\bar{B}_{n-2})}.$$

The further calculation of the functions $t(A \mid B)$ and $v(A \mid B)$, in which we may restrict ourselves to primary classes, will be carried out only later.

§ 9. Representation of the function $\bar{\gamma}$ by t-functions and of $\bar{\delta}$ by v-functions.

Having shown in § 8 how the functions t and v can be expressed by ψ-functions, we now show how the functions $\bar{\gamma}$ and $\bar{\delta}$ can be represented with the aid of t and v.